

# Appendix A. Stochastic conformance checking with guarantees – Proofs

This document contains formal proofs that justify the proposed stochastic conformance checking approach.

**Proposition 1** (Well-Formedness). *The intersection of two well-formed FLMCs is a well-formed FLMC.*

*Proof.* By construction, the intersection model contains only reachable states, as these correspond to pairs of reachable states of the original models. Dead states are removed according to the definition of intersection. The resulting model is also split by construction, since transitions on different symbols lead to different state pairs, as they lead to different states in the original models. Finally, the empty sequence is not accepted by the intersection model, since it is not accepted by either of the original models.  $\square$

**Property 2** (Representation). *A stochastic-aware conformance measure should depend on the stochastic languages of logs and models and not on their representations.*

*Proof.* Different FLMC representations of the same stochastic language may contain several equivalent states that correspond to the same stochastic language generated when starting in these states. Such states can be merged without changing the represented stochastic language.

Let  $s_1, \dots, s_k$  be equivalent states that are merged into a single state  $s$ . The stationary probability of  $s$  in the merged FLMC equals the sum of their stationary probabilities:

$$\pi(s) = \sum_{i=1}^k \pi(s_i).$$

Hence, when either of the models with states  $s_1, \dots, s_k$  or single state  $s$  intersects with another model with a corresponding state  $r$ , for the conformance measures the value of

$$\pi(r) - \sum \pi(r, s_i),$$

equals

$$\pi(r) - \pi(r, s).$$

Therefore, merging equivalent states does not change the precision and recall measures. Conversely, splitting a state into equivalent states does not change it either.  $\square$

Below, we present proofs of the monotonicity of the steady-state approach.

**Lemma 1** (Stationary probabilities and trace executions). *Let  $MC = (S, s_0, F, \Sigma, \text{prob}, \text{trans})$  be an accepting well-formed FLMC representing a stochastic language  $\mathcal{S}$ , and let  $N_s(t)$  denote the number of visits of state  $s$  during execution of trace  $t$ . The normalized stationary probability of every non-initial state  $s \in S$  is determined as:*

$$\frac{\pi(s)}{1 - \pi(s_0)} = \frac{\sum_{t \in \text{dom}(\mathcal{S})} \mathcal{S}(t) N_s(t)}{\sum_{r \in S \setminus \{s_0\}} \sum_{t \in \text{dom}(\mathcal{S})} \mathcal{S}(t) N_r(t)}.$$

*Proof.* Since  $MC$  is accepting, every completed trace reaches an accepting state and then returns to the initial state  $s_0$  through a  $\chi$ -transition. Hence, each return to  $s_0$  defines the end of one execution cycle and the beginning of the next one.

The expected number of visits to  $s$  in one cycle is  $\mathbb{E}[N_s] = \sum_{t \in \text{dom}(\mathcal{S})} \mathcal{S}(t) N_s(t)$ .

The expected number of visits for all states except the initial state is  $\mathbb{E}[N] = \sum_{r \in S \setminus s_0} \sum_{t \in \text{dom}(\mathcal{S})} \mathcal{S}(t) N_r(t)$ .

$$\frac{\mathbb{E}[N_s]}{\mathbb{E}[N]} = \frac{\sum_{t \in \text{dom}(\mathcal{S})} \mathcal{S}(t) N_s(t)}{\sum_{r \in S \setminus \{s_0\}} \sum_{t \in \text{dom}(\mathcal{S})} \mathcal{S}(t) N_r(t)}.$$

Since the FLMC is split, transitions carrying different labels cannot connect the same pair of states. Hence, visits to states can be associated with the corresponding labelled trace executions without conflating different labelled transitions.  $\square$

**Property 11** (Whole-model monotonicity of the intersection). *Let  $MC_M$ ,  $MC_{L_1}$ , and  $MC_{L_2}$  be accepting well-formed FLMCs representing  $\mathcal{S}_M$ ,  $\mathcal{S}_{L_1}$ , and  $\mathcal{S}_{L_2}$ , respectively, and let  $MC_{I_i} = MC_M \cap MC_{L_i}$ ,  $i \in 1, 2$ .*

*Let*

$$p_{I_i}(s_M) = \frac{\sum_{(s_M, s_{L_i}) \text{ is reachable}} \pi(s_M, s_{L_i})}{1 - \pi(s_{0M}, s_{0L_i})}.$$

*If  $|\mathcal{S}_{L_1}(t) - \mathcal{S}_M(t)| \leq |\mathcal{S}_{L_2}(t) - \mathcal{S}_M(t)|$  for every  $t \in \text{dom}(\mathcal{S}_M)$ , then*

$$\sum_{s_M \neq s_{0M}} \max\left(\frac{\pi(s_M)}{1 - \pi(s_{0M})} - p_{I_1}(s_M), 0\right) \leq \sum_{s_M \neq s_{0M}} \max\left(\frac{\pi(s_M)}{1 - \pi(s_{0M})} - p_{I_2}(s_M), 0\right).$$

*Proof.* By the preceding lemma, the normalized stationary probability of every non-initial model state is determined by the probability mass of the traces whose executions visit that state.

The intersection retains only common labelled behaviour. For every common transition, the unnormalized probability assigned by the intersection is  $\min(\text{prob}_M(s_M, a), \text{prob}_{L_i}(s_{L_i}, a))$ . Thus, the minimum operation retains the probability mass shared by the model and the corresponding log and removes probability mass that is not shared.

The discrepancies are not considered state by state in isolation. Since stochastic languages are probability distributions,  $\sum_t \mathcal{S}_M(t) = \sum_t \mathcal{S}_{L_i}(t) = 1$ :

$$\sum_t \max(\mathcal{S}_M(t) - \mathcal{S}_{L_i}(t), 0) = \frac{1}{2} \sum_t |\mathcal{S}_M(t) - \mathcal{S}_{L_i}(t)|.$$

Hence, an excess of probability relative to the model for some traces must be compensated by a deficit for other traces. Therefore, although the minimum operation clips an excess locally, the corresponding probability discrepancy is necessarily represented by a deficit elsewhere in the complete distribution.

By assumption,  $|\mathcal{S}_{L_1}(t) - \mathcal{S}_M(t)| \leq |\mathcal{S}_{L_2}(t) - \mathcal{S}_M(t)|$  for every  $t \in \text{dom}(\mathcal{S}_M)$ . Consequently,

$$\sum_t |\mathcal{S}_{L_1}(t) - \mathcal{S}_M(t)| \leq \sum_t |\mathcal{S}_{L_2}(t) - \mathcal{S}_M(t)|.$$

Thus, the total probability mass not shared with the model cannot be larger for  $\mathcal{S}_{L_1}$  than for  $\mathcal{S}_{L_2}$ .

Since the FLMCs are split, the shared probability mass is propagated through distinct labelled transitions and can be associated with the corresponding model trace prefixes. By the preceding lemma, the probabilities of these trace executions determine the normalized stationary probabilities. Moreover, the precision measure sums the stationary probabilities of all reachable intersection states corresponding to each model state. Therefore, probability mass distributed among several intersection states corresponding to the same model state is aggregated before the discrepancy is calculated.

It follows that the total positive stationary-probability deficit of the intersection with respect to the model cannot increase when the stochastic language is moved closer to the model:

$$\sum_{s_M \neq s_{0M}} \max\left(\frac{\pi(s_M)}{1 - \pi(s_{0M})} - p_{I_1}(s_M), 0\right) \leq \sum_{s_M \neq s_{0M}} \max\left(\frac{\pi(s_M)}{1 - \pi(s_{0M})} - p_{I_2}(s_M), 0\right).$$

□